



OPEN Knowledge, economic, and WASH challenges affecting menstrual health and hygiene among nursing students in Bangladesh using a mixed-methods approach

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Menstrual health and hygiene (MHH) remain stigmatized and neglected in low- and middle-income countries such as Bangladesh due to cultural taboos, poor infrastructure, and limited education. This study examines how knowledge, socioeconomic background, and WASH facilities are associated with key MHH indicators. A convergent mixed-methods design was applied between December 2024 and January 2025, combining a survey of 370 students from seven public nursing institutions with six focus group discussions involving 21 female and 21 male students for gendered understanding. Quantitative data were analyzed using descriptive statistics and multivariable logistic regression, while qualitative data were examined through thematic analysis. Quantitative results showed that adequate access to menstrual products was associated with higher odds among students enrolled in the B.Sc. in Nursing program (AOR = 6.91, 95% CI 2.44–19.55, $p < 0.05$), as well as among students whose mothers had secondary education (AOR = 4.82, 95% CI 1.52–15.28, $p < 0.05$). Adequate WASH facilities were also associated with higher odds among students in the B.Sc. in Midwifery (Post Basic) program (AOR = 2.87, 95% CI 1.28–6.43, $p < 0.05$) and those whose mothers had higher secondary education (AOR = 6.19, 95% CI 1.47–26.00, $p < 0.05$). Qualitative findings revealed limited menstrual knowledge prior to menarche, embarrassment in purchasing products, unmanaged menstrual pain, and inadequate privacy and disposal facilities within campuses. Male students reported minimal formal exposure to menstrual education but demonstrated increased awareness of menstrual pain, privacy needs, and disposal challenges after engaging in shared academic and residential environments. This study is novel in integrating male student perspectives with institutional WASH constraints in nursing education, positioning menstruation as both a gendered and systems-level issue. The findings support actionable recommendations for gender-inclusive MHH curricula, minimum WASH standards in educational institutions, and policy alignment between nursing education, sanitation infrastructure, and menstrual waste management to promote menstrual dignity and equitable health training.

Keywords Menstrual health and hygiene, Gendered perspectives, Students, Mixed-methods, Bangladesh

Menstrual Health and Hygiene (MHH) are increasingly recognized as a critical component of public health, gender equity, and human rights¹. Despite growing international advocacy and policy attention, menstruation remains shrouded in stigma, misinformation, and infrastructural neglect. This is especially true in Low-and Middle-Income Countries (LMICs), where access to safe, affordable, and culturally appropriate menstrual materials and sanitation facilities is often limited^{2,3}. In this context, MHH indicators such as access to adequate menstrual materials, knowledge and education, Water, Sanitation and Hygiene (WASH) facilities, social support, and the ability to manage pain and emotional discomfort serve as measurable benchmarks to assess the quality and inclusiveness of menstrual health systems⁴. In South Asia, including Bangladesh, these challenges are

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compounded by deeply entrenched social norms that frame menstruation as a source of shame and impurity, contributing to inadequate knowledge, restricted mobility, and poor menstrual hygiene practices^{5–7}.

As Bangladesh is a Muslim-majority country, everyday interpretations of Islam often influence beliefs around menstrual purity and modesty, which can reinforce silence, stigma, and restrictions on discussing menstruation openly (e.g., limited pre-menarche knowledge and stigma around menstrual blood)⁸. To address these issues, the government introduced several policy reforms, including the incorporation of MHH into the 2016 National Curriculum and the adoption of the Water and Sanitation Sector Development Plan (2011–2025). However, despite these policy commitments, implementation has been inconsistent, with gaps in institutional coverage, infrastructure delivery, and monitoring mechanisms^{9,10}. Recent studies have begun to illuminate the emotional and practical burdens that menstruators face, including fear and confusion at menarche, discomfort in accessing products, and the absence of safe, private spaces for menstrual management^{11,12}. However, much of the related research has focused narrowly on adolescent schoolgirls, with limited attention given to young adult women in health-related disciplines, particularly nursing students. These students play a dual role, not only as adult learners but also future healthcare providers responsible for delivering sexual and reproductive health education and care to the wider community. Evidence suggests that despite formal health training, many nursing students continue to experience embarrassment, misinformation, inadequate facilities, and emotional distress related to menstruation, alongside variations in product use and disposal practices^{5,13}.

Gender dynamics further complicate menstrual experiences. Female students often report secrecy, emotional burdens, and social restrictions, suggesting that formal education alone is insufficient to dismantle deeply rooted menstrual stigma^{13,14}. Although male students are not biologically affected by menstruation, they often lack menstrual education and may hold stigmatizing views due to societal silence on the topic^{15,16}. Investigating these gendered perspectives is essential for designing gender-sensitive interventions that can transform not only knowledge but also attitudes and behaviours among future healthcare providers.

This study aims to examine how knowledge, socioeconomic factors, and WASH conditions shape MHH among nursing students in Bangladesh. It contributes to ongoing debates on menstrual stigma, gendered health education, and the adequacy of MHH integration within professional healthcare training in low-resource settings. By adopting a mixed-methods approach and incorporating both female and male nursing students, the study expands the scope of existing research beyond adolescent populations and foregrounds the role of future healthcare providers as both menstruators and agents of change. The findings provide evidence to inform culturally responsive MHH policies, curriculum reform, and improvements in institutional WASH infrastructure within nursing education and comparable health training contexts.

Methods and materials

Study design and rationale

This study adopted a convergent parallel mixed-methods design to comprehensively examine MHH practices, experiences, and gendered perceptions among nursing and midwifery students in Bangladesh. The use of both quantitative and qualitative approaches was deliberately aligned with the study aims. The quantitative component enabled measurement of the prevalence and distribution of key MHH indicators and their sociodemographic determinants, while the qualitative component provided contextualized insights into gendered meanings, stigma, coping strategies, and sociocultural constraints surrounding menstruation that cannot be adequately captured through survey data alone. Quantitative and qualitative data were collected concurrently, analyzed independently, and integrated during interpretation to allow triangulation and complementarity of findings, thereby strengthening the validity and depth of the conclusions¹⁷. This design was particularly appropriate given the sensitive, socially embedded, and gendered nature of menstruation within a resource-constrained and gender-segregated educational context.

Theoretical and conceptual framework

To guide measurement and interpretation of stigma and gender norms, we used an integrated conceptual framework drawing on menstrual stigma theory, gender norms theory, and the socio-ecological model of health. Menstrual stigma theory conceptualizes menstruation as a socially constructed and morally regulated process, maintained through secrecy, shame, and silence, which restricts information flow and help-seeking behaviors^{18,19}. Gender norms theory further explains how menstruation is feminized and excluded from male socialization, positioning men as uninformed bystanders while placing the burden of concealment and self-management on women^{4,20}. These norms are reinforced within families, educational institutions, and healthcare training environments. The socio-ecological model was used to situate stigma and gender norms across interacting levels: individual (knowledge, pain management), interpersonal (maternal education, peer and male responses), institutional (curriculum gaps, WASH facilities), and structural (socioeconomic status, rural–urban inequities)²¹. This framework informed the mixed-methods design by guiding variable selection for the quantitative component and providing a structure for qualitative probes exploring stigma, silence, and gendered responsibility. By integrating these perspectives, the study conceptualizes menstrual health not merely as an individual hygiene practice but as a gendered and structurally mediated health experience, helping to explain observed contradictions related to knowledge, socioeconomics and WASH challenges.

Study setting and population

The study was conducted across seven government nursing and midwifery colleges located in four districts, Dhaka, Rajshahi, Naogaon, and Satkhira, in three out of the eight regions in Bangladesh. These sites were strategically chosen to ensure representation from different administrative divisions (Dhaka as capital city, Rajshahi, and Khulna), providing diversity in urban and rural, socioeconomic, infrastructural, and cultural contexts. Each district in the selected regions has a government nursing college. Seven of the thirty-eight colleges

were purposively selected to represent nursing students from diverse backgrounds. Government nursing colleges were chosen because they are more accessible for researcher and provide an appropriate environment for exploring sensitive issues such as menstruation among adults. The study population consisted of full-time students aged 18 years and older who were enrolled in accredited nursing and midwifery programs and had experienced menarche. Participants were eligible if they had menstruated within the past three months for quantitative data, ensuring the relevance of self-reported menstrual experiences. Exclusion criteria included students who were under 18 years of age, unwilling to provide informed consent, pregnant, or had not menstruated within the previous three months. For the qualitative component, both male and female students were included to capture a holistic, gendered understanding of MHH.

Sampling strategy

A total of 370 female participants were recruited using stratified random sampling for quantitative data. Strata were created on the basis of institution, year of study, and academic discipline (nursing or midwifery). Proportional allocation ensures that each stratum contributes to the final sample in accordance with its population size. The sample size was calculated via Cochran's formula for estimating proportions²²:

$$n_o = \frac{z^2 \times pq}{e^2} \quad (1)$$

where n_o = is the initial sample size, $z = 1.96$ for a confidence level of 95%, $p = 0.50$ (expressed as 50% in decimal form), $q = 1 - p$, and $e = 0.05$ (representing a 5% margin of error). An assumed prevalence of 50% was used to maximize variability and obtain a conservative estimate in the absence of reliable, comparable prevalence data across multiple MHH indicators among nursing and midwifery students in Bangladesh. Using these parameters, the initial sample size was 384. As the total population of eligible female students across the selected seven government nursing and midwifery colleges was finite ($N=2,700$), a finite population correction (FPC) was applied using the standard formula:

$$n = \frac{n_o}{[1 + \{(n_o - 1)/N\}]} \quad (2)$$

After applying the FPC and allowing for an anticipated 10% nonresponse rate, the final target sample size was 370 participants.

For the qualitative strand, 42 participants (21 males and 21 females) were selected using purposive sampling to capture diversity in academic year, socioeconomic background, and geographic origin. Participants were recruited from all seven institutes, with approximately six participants selected from each institute (three males and three females). This approach was used to ensure balanced institutional representation rather than strict proportional allocation, given the exploratory and interpretive aims of the qualitative component. Participants were organized into six FGDs, three male-only and three female-only groups, with seven participants per group. This group size was chosen to facilitate active interaction, allow diverse viewpoints to emerge, and maintain manageable group dynamics for in-depth discussion, consistent with established qualitative research guidance²³ (Fig. 1).

Data collection procedures

The structured survey was developed by adapting the shortlist of priority MHH indicators²⁴, which was specifically designed to standardize national and global MHH monitoring [Supplementary file 1]. The questionnaire covered seven domains, including menstrual materials with two items, WASH facilities with two items, menstrual knowledge with three items, discomforts and disorders with one item, supportive social environments with one item, menstrual health impacts with one item, and policies with one item, comprising a total of 11 questions for each participant. The tool was translated into Bangla, followed by back-translation to ensure conceptual equivalence. To ensure content validity, the instrument was reviewed by three experts in public health nursing and women's health. It was pilot tested with 30 nursing students from similar institutions to assess clarity and reliability. The scale showed good internal consistency (Cronbach's $\alpha = 0.83$). The data were collected through face-to-face interviews conducted by the first author and two trained research assistants, both qualified nurses and midwives with prior experience in survey-based research. Before data collection, all data collectors received structured training on the study objectives, questionnaire content, interview techniques, ethical conduct, confidentiality, and procedures for obtaining informed consent. Particular emphasis was placed on maintaining privacy, minimizing participant discomfort, and handling menstruation related discussions sensitively. Data were collected in private rooms within college premises to ensure confidentiality and participant comfort. The data collection period spanned from 01 December 2024 to 31 January 2025.

The FGD guides were developed on the basis of previous studies in similar contexts^{13,25} [Supplementary file 2]. The female FGDs explored firsthand menstrual experiences, product choices, emotional burdens, social taboos, and coping strategies. In contrast, the male FGDs focused on perceptions, misinformation, social attitudes, and readiness to support peers. All FGDs were conducted in Bangla, audio-recorded with participants' consent, and lasted approximately 70–90 min. Sessions were facilitated by gender-matched moderators: female facilitators for female FGDs and the first author for the male to minimize response bias and foster participant comfort. The transcripts were transcribed verbatim and subsequently translated into English for analysis.

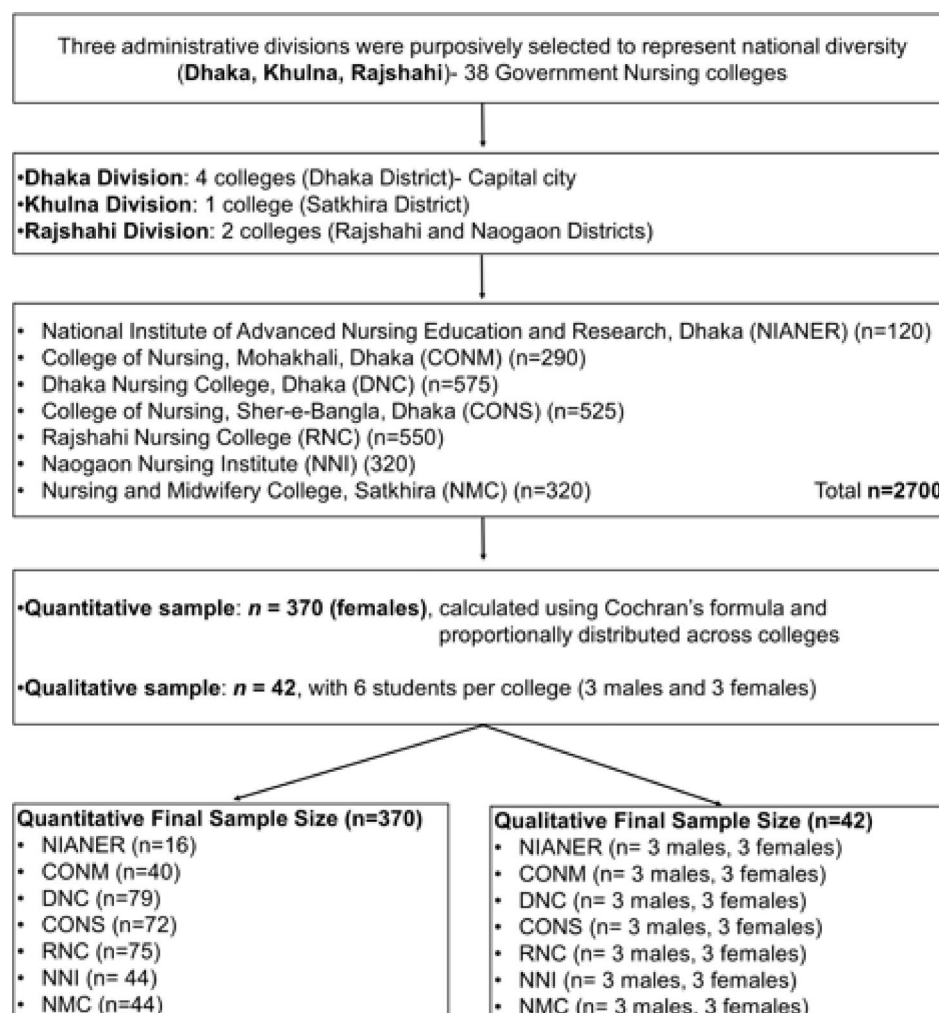


Fig. 1. Flowchart for participants selection process.

Independent variables

Sociodemographic characteristics were treated as independent variables in the multivariable logistic regression analyses. These included academic program (Diploma in Midwifery [reference category], Diploma in Nursing and Midwifery, B.Sc. in Midwifery, B.Sc. in Nursing, and MSc in Nursing), year of study (first year [reference], second year, third year, and fourth year), father's educational attainment (graduate or above [reference], and no formal education), mother's educational attainment (graduate or above [reference], higher secondary, secondary, primary, and no formal education), birthplace status (rural [reference] or urban), and monthly family income (below BDT 10,000 [reference], BDT 10,000–20,000, BDT 20,000–30,000, and above BDT 30,000). These variables were selected a priori based on existing evidence linking educational exposure, socioeconomic position, and place-based inequities with MHH outcomes and were entered into the regression models as categorical predictors.

Outcome variables

The dependent variables consisted of six MHH indicators derived from the priority MHH indicator shortlist tool and were analyzed as binary outcomes in separate logistic regression models. Adequate menstrual materials were defined as having sufficient menstrual materials to change as often as needed during the last menstrual period and was coded as 1 for “yes” and 0 for “no” or “do not remember.” Adequate WASH facilities reflected conditions supportive of safe and dignified changing of menstrual materials at school, based on reported cleanliness and perceived safety/privacy of the changing location; responses indicating adequate conditions were coded as 1, while inadequate or uncertain responses were coded as 0. Adequate menstrual knowledge captured sufficient menstrual-related knowledge or education exposure (including knowledge prior to menarche or correct identification of the fertile period) and was dichotomized as adequate (1) versus inadequate or uncertain (0). Adequate menstrual health impact was defined to capture functional consequences of menstruation on students' daily and academic activities. This composite outcome was constructed using items assessing whether menstruation negatively affected daily activities and participation in class, as both reflect functional impairment within the educational context. Participants reporting no negative impact were coded as 1, and those reporting

any negative impact were coded as 0. Supporting social environment represented access to interpersonal or professional support for menstrual concerns (e.g., ability to seek help from a health worker or trusted person) and was coded as 1 for “yes” and 0 for “no”; responses indicating “not needed” or uncertainty were treated as inadequate and coded as 0 to maintain analytical consistency. Finally, increased discomfort was used as a negative outcome and defined as inability to reduce menstrual pain when needed; participants reporting increased discomfort were coded as 1, while those reporting no increased discomfort were coded as 0. This harmonized dichotomization allowed consistent interpretation of adjusted odds ratios across all regression models.

Data analysis

Quantitative data were entered and analyzed via JMP Student Version 18 (SAS Institute Inc., USA). Descriptive statistics, including frequencies, percentages, means, standard deviations, were calculated for all the variables. Associations between sociodemographic characteristics (independent variables) and MHH indicators (dependent variables) were initially examined using unadjusted (simple) logistic regression to estimate crude odds ratios. For indicators assessing menstrual materials, WASH facilities, menstrual knowledge, menstrual health impact, and the supporting social environment, responses were dichotomized, with “yes” indicating adequate and coded as 1, and “no” or “do not remember” indicating inadequate and coded as 0. For the discomfort model, “increased discomfort” was coded as 1, and the absence of increased discomfort was coded as 0. This coding approach ensured consistent interpretation of adjusted odds ratios across models. To control for potential confounders, variables with $p < 0.20$ in bivariate analyses were entered into the multivariable models, in accordance with recommended epidemiological model-building practices to avoid the premature exclusion of relevant covariates and to improve model stability²⁶. For multivariable analysis, separate binary logistic regression models were constructed for each MHH indicator and sociodemographic characteristic. Adjusted odds ratios (AORs) with 95% confidence intervals (CIs) are reported. Statistical significance in the final models was defined as $p < 0.05$.

Qualitative data were analyzed via thematic analysis guided by Braun and Clarke’s six-phase approach^{27,28}. The analysis process included becoming familiar with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the final report. Coding was performed both inductively (from data) and deductively (from the FGD guide). For example, participants’ descriptions of hiding menstrual products or avoiding male peers were initially coded as “concealment” and “fear of exposure”, which were later grouped under the broader theme of Menstrual product accessibility and availability. The transcripts were imported into MAXQDA Analytics Pro 2024 for coding and organization. Coding was performed independently by three researchers to enhance intercoder reliability, with discrepancies resolved through consensus. The themes were further validated through peer debriefing, and representative direct quotes were selected to illustrate thematic findings. Data from male and female FGDs were analyzed separately to preserve gendered narratives and then compared for points of convergence and divergence.

Reflexivity and positionality

The research team comprised male and female researchers with backgrounds in nursing, public health, human ecology, and qualitative methods. The first author is a male registered nurse from Bangladesh with extensive experience in MHH research. His professional training and cultural familiarity with the study context facilitated rapport with institutional stakeholders and informed the framing of research questions; however, his gender and professional status may have influenced how female participants initially articulated sensitive menstrual experiences. To mitigate this, female research assistants, trained nurses and midwives conducted focus group discussions and interviews with female participants, which helped reduce gendered power imbalances and encouraged more open dialogue. The research assistants shared professional identity with participants may have fostered trust but also risked assumptions of shared knowledge or normalization of menstrual challenges. To address this, facilitators used open-ended prompts and avoided affirming participants’ responses during discussions. Throughout the study, the team engaged in reflexive journaling to critically examine how personal assumptions, professional roles, and cultural norms shaped data collection and interpretation. Regular peer debriefings and audit trails were used to challenge interpretations and enhance analytical rigor and trustworthiness²⁹.

Ethical considerations and consent to participate

Ethical approval was obtained from the Ethical Review Committee, Faculty of Health Sciences, Hokkaido University (Ref: 23–52), Directorate General of Nursing and Midwifery, Bangladesh (Ref: 45.03.0000.008.14.097.2021–650) and the National Institute of Advanced Nursing Education and Research, Bangladesh (Ref: NIA-OS-2023-09). All methods were performed in accordance with the relevant guidelines and regulations, including the Declaration of Helsinki and applicable institutional ethical standards. All participants provided written informed consent after receiving detailed information about the study objectives, procedures, voluntary nature of participation, confidentiality measures, and the right to withdraw at any time without consequences. To acknowledge participants’ time and transportation costs, a small non-coercive token of appreciation (refreshments and stationery items) was provided in line with local ethical guidelines; no monetary incentives were offered. Confidentiality was maintained by anonymizing the data and securing all files digitally with password protection.

Results

Sociodemographic characteristics of the quantitative findings

The mean age of the participants was 23.7 years ($SD \pm 3.1$), and the mean age at menarche was 13 years ($SD \pm 4.21$). The majority of participants were nursing students (60%), while 40% were enrolled in midwifery programs. With respect to course enrollment, over half were pursuing a diploma in midwifery (56.5%), followed by the B. Sc in Nursing (4 years) program (14.3%) and MSc in Nursing (9.4%). A smaller proportion was

enrolled in the Diploma in Nursing & Midwifery (5.9%), B. Sc in Midwifery (Post Basic) (8.65%), and B. Sc in Nursing (Post Basic) (5.1%) programs. Most participants were in their first year (47.5%), with decreasing representation across the second (20.81%), third (26.2%), and fourth years (5.4%). In terms of religion, the majority identified as Muslim (88.6%), followed by Hindu (9.4%), Christian (1.3%), and Buddhist (0.5%). A substantial proportion of participants reported that their fathers had completed primary education (32.9%) or secondary education (28.9%), whereas only 12.4% of fathers had attained graduate-level education or higher. Mothers' education levels were generally lower, with only 5.1% reporting graduate-level education or above; the majority had completed primary (39.7%) or secondary education (36.4%). Most participants were born in rural areas (79.7%) and resided in hostels during their studies (72.9%). In terms of economic status, 30.5% of the participants reported a monthly family income below BDT 10,000, whereas only 17.8% reported a monthly income exceeding BDT 30,000 (Table 1).

Menstrual health and hygiene (MHH) indicators

A majority of the participants (71.3%) reported having enough menstrual materials to change as often as needed during their last menstrual period, while 20.2% did not, and 8.3% could not recall them. Less than half (41.8%) changed their menstrual materials at school during their last period, indicating limited access to appropriate facilities. While 43.2% stated that the place they used to change was clean, over one-third (35.6%) reported unclean conditions. Privacy concerns were notable; 29.7% of the participants were worried about being seen while changing, whereas 47.8% said they felt safe during the process. Regarding menstrual education, 47.8% had received information during secondary school, 3.7% in primary school, and 28.6% during both educational

Variables	Category	n (%) or Mean $\pm$ SD
Age		23.7 $\pm$ 3.1
Age at menarche		13 $\pm$ 4.2
Category of studentship	Nursing	222 (60)
	Midwifery	148 (40)
Name of course	Diploma in Midwifery	209 (56.5)
	Diploma in Nursing & Midwifery	22 (5.9)
	B.Sc in Midwifery (Post Basic)	32 (8.6)
	B.Sc in Nursing (Post Basic)	19 (5.2)
	B.Sc in Nursing (4 years)	53 (14.3)
	MSc in Nursing	35 (9.4)
Year of college	First year	176 (47.5)
	Second year	77 (20.8)
	Third year	97 (26.2)
	Fourth year	20 (5.4)
Religion	Islam	328 (88.6)
	Hinduism	35 (9.4)
	Buddhism	2 (0.5)
	Christian	5 (1.3)
Fathers education level	Graduate or above	46 (12.4)
	Higher Secondary	58 (15.6)
	Secondary	107 (28.9)
	Primary	122 (32.9)
	Illiterate	37 (10)
Mother's education level	Graduate or above	19 (5.14)
	Higher Secondary	38 (10.2)
	Secondary	135 (36.4)
	Primary	147 (39.7)
	Illiterate	31 (8.3)
Birthplace status	Urban	75 (20.2)
	Rural	295 (79.7)
Resident status	Hostel	270 (72.9)
	Own house	100 (27)
Monthly family income	Below 10,000 BDT*	113 (30.5)
	10,000–20,000 BDT	111 (30)
	20,000–30,000 BDT	80 (21.6)
	More than 30,000 BDT	66 (17.8)

Table 1. Sociodemographic characteristics of the quantitative findings ($N=370$). *BDT= Bangladeshi Taka.

periods. However, 19.7% reported receiving no menstrual education in school. Notably, only 45.1% had knowledge about menstruation prior to their first period, whereas 49.7% did not. Knowledge about fertility was moderate, with 61.3% accurately identifying the most fertile time in the menstrual cycle. In contrast, 15.1% responded inaccurately, and 23.5% were uncertain. Menstruation disrupted daily activities for 45.4% of the students, and 58.6% experienced difficulty participating in class during menstruation, highlighting a significant burden on academic engagement and wellbeing. Similarly, 71.3% had someone to approach for advice, emotional support, or resources, 19.4% lacked such support, and 9.1% reported that such support was not needed. When addressing menstrual pain management, 45.4% successfully reduced discomfort, whereas 23.5% could not, and 31% could not recall it. Most participants (74%) felt comfortable seeking support from healthcare providers regarding menstruation (Table 2).

Significant sociodemographic characteristics with MHH indicators

Multivariable logistic regression analysis identified several significant associations between sociodemographic characteristics and MHH indicators (Table 3). For all adequacy outcomes, adjusted odds ratios (AORs > 1) indicate higher odds of meeting the predefined adequacy criteria, whereas for the increased discomfort outcome, AORs > 1 indicate higher odds of experiencing greater menstrual discomfort. Using students enrolled in the Diploma in Midwifery program as the reference group, those enrolled in the B.Sc. in Nursing (4 years) program had significantly higher odds of reporting access to adequate menstrual materials (AOR = 6.91, 95% CI: 2.44–19.55, $p < 0.05$). Similarly, students enrolled in the MSc in Nursing program also had significantly higher odds of reporting access to adequate menstrual material compared with the reference group (AOR = 2.94, 95% CI: 1.05–8.21, $p < 0.05$). With respect to adequate WASH facilities, students in the B.Sc. in Midwifery (Post Basic) program (AOR = 2.87, 95% CI: 1.28–6.43, $p < 0.05$) and those in the Diploma in Nursing and Midwifery program (AOR = 2.72, 95% CI: 1.01–7.32, $p < 0.05$) had higher odds compared with the reference group.

Year of study was also significantly associated with MHH indicators. Second- and third-year students had higher odds of reporting adequate WASH facilities (second year: AOR = 2.27, 95% CI: 1.22–4.23, $p < 0.05$;

Indicator	Variables	Category	n (%)
Menstrual materials	Had enough menstrual materials to change as often as needed during the last period	Yes	264 (71.3)
		No	75 (20.2)
		Do not remember	31 (8.3)
WASH facilities	Changed menstrual materials at school during the last period	Yes	155 (41.8)
		No	215 (58.1)
	The place for changing menstrual materials at school was clean	Yes	160 (43.2)
		No	132 (35.6)
	Felt worried that someone might see while changing at school	Yes	110 (29.7)
		No	179 (48.3)
	Felt safe while changing menstrual materials at school	Yes	177 (47.8)
		No	104 (28.1)
Menstrual knowledge	Received education about menstruation in school	Primary	14 (3.7)
		Secondary	177 (47.8)
		Both	106 (28.6)
		None	73 (19.7)
	Knew about menstruation before the first period	Yes	167 (45.1)
		No	184 (49.7)
		Do not remember	19 (5.1)
	Knows the most fertile time in the menstrual cycle	Yes	227 (61.3)
		No	56 (15.1)
		Do not remember	87 (23.5)
Menstrual health impact	Menstruation does not impact daily activities	Yes	202 (54.5)
		No	168 (45.4)
	Had difficulty participating in class due to menstruation	Yes	217 (58.6)
		No	153 (41.3)
Social environment	Has someone to ask for support (advice, resources, or emotional help) during menstruation	Yes	264 (71.3)
		No	72 (19.4)
		Not needed	34 (9.1)
Discomfort	Able to reduce menstrual pain during the last period when needed	Yes	168 (45.4)
		No	87 (23.5)
		Do not remember	115 (31)
	Feels comfortable seeking help from a health worker for menstrual concerns	Yes	274 (74)
		No	96 (25.9)

Table 2. Menstrual health and hygiene indicators (N = 370).

third year: AOR=1.92, 95% CI: 1.03–3.57, $p<0.05$), and fourth-year students demonstrated even higher odds (AOR=3.18, 95% CI: 1.04–9.68, $p<0.05$). However, fourth-year students had significantly higher odds of reporting access to adequate menstrual materials (AOR=2.18, 95% CI: 1.01–3.31, $p<0.05$), potentially reflecting cumulative educational exposure.

Parental educational levels, particularly maternal education, showed consistent associations with multiple MHH indicators. Compared with students whose mothers had graduate-level education, those whose mothers had secondary education had higher odds of reporting adequate menstrual materials (AOR=4.82, 95% CI: 1.52–15.28, $p<0.05$) and adequate WASH facilities (AOR=5.06, 95% CI: 1.36–18.85, $p<0.05$). However, these estimates were characterized by wide confidence intervals, indicating statistical imprecision and potentially limited power within maternal education subgroups.

For the discomfort model, where increased discomfort was coded as the outcome, primary level maternal education (AOR=2.17, 95% CI: 1.18–3.99, $p<0.05$), and no formal education levels (AOR=2.26, 95% CI: 1.21–4.21, $p<0.05$) were significantly associated with higher odds of increased discomfort compared with graduate or above maternal education, indicating a protective effect of higher maternal education. Additionally, students whose fathers had no formal education had higher odds of reporting increased menstrual health problems (AOR=2.26, 95% CI: 1.10–5.23, $p<0.05$), which aligns with qualitative findings indicating limited paternal involvements in MHH education and practices.

Geographical and economic factors were also associated with MHH outcomes. Students from urban areas had higher odds of reporting adequate menstrual knowledge (AOR=1.52, 95% CI: 1.30–3.91, $p<0.05$) and adequate menstrual health impact (AOR=1.48, 95% CI: 1.27–5.83, $p<0.05$). Compared with students from households earning less than BDT 10,000 per month, those from households earning more than BDT 30,000 per month had higher odds of reporting an adequate supporting social environment (AOR=1.44, 95% CI: 1.22–2.87, $p<0.05$).

Qualitative perspectives on menstrual health and hygiene

The qualitative findings provide contextual depth to the quantitative results by explaining how and why observed disparities in MHH indicators occur. This section presents a detailed synthesis of the qualitative findings derived from FGDs with both female and male nursing students. The FGDs included 42 nursing students, equally divided by sex (21 males and 21 females), with a mean age of 23.1 years (SD=3). The participants were enrolled in MSc in nursing (24%), B. Sc in Nursing (33%), and B. Sc. in midwifery (Post Basic) programs (43%). The majority identified as Muslim (81%), whereas some were Hindu (19%). Most participants were from urban areas (60%) and were single (64%). Monthly family income varied, with over half (55%) earning between 20,000 and 30,000 BDT. With respect to clinical experience, half had 2–4 years of hospital exposure, whereas 29% had more than 4 years (Table 4).

Thematic analysis yielded four overarching themes and thirteen subthemes that illuminate the sociocultural, emotional, and infrastructural dimensions of MHH and are listed in Table 5. In the FGDs, individual participants were identified via the labels Male (M), Female (F) and the Participants’ Number (e.g., P1, P2) to represent each person.

Menstrual knowledge and information

Consistent with the quantitative finding that only 45.1% of students had knowledge of menstruation before menarche (Table 2) and that maternal primary education level had higher odds of participants’ adequate menstrual knowledge (Table 3; AOR=1.35), qualitative narratives revealed widespread lack of early education, reliance on mothers or sisters, and exclusion of male students from menstrual information.

Variables		Male (N=21) n (%) or M ± SD	Female (N=21) n (%) or M ± SD	Total (N=42) n (%) or M ± SD
Age		23.2 ± 3.5	23.1 ± 2.7	23.1 ± 3
Course name	M.Sc. in Nursing	4 (19)	6 (29)	10 (24)
	B.Sc. in Nursing	6 (29)	8 (38)	14 (33)
	B.Sc. in Midwifery (Post Basic)	11 (52)	7 (33)	18 (43)
Religion	Islam	18 (86)	16 (76)	34 (81)
	Hinduism	3 (14)	5 (24)	8 (19)
Birthplace	Urban	12 (57)	13 (62)	25 (60)
	Rural	9 (43)	8 (38)	17 (40)
Marital status	Single	12 (57)	15 (71)	27 (64)
	Married	9 (43)	6 (29)	15 (36)
Monthly family income	10,000–20,000/-	4 (19)	5 (24)	9 (21)
	20,000–30,000/-	11 (52)	12 (57)	23 (55)
	More than 30,000/-	6 (29)	4 (19)	10 (24)
Clinical experience in hospital	Below 2 years	3 (14)	6 (29)	9 (21)
	2–4 years	10 (48)	11 (52)	21 (50)
	Above 4 years	8 (38)	4 (19)	12 (29)

Table 4. Characteristics of the focus group discussion participants (N=42).

Theme	Subtheme
Menstrual Knowledge and Information	1.1 Lack of early education 1.2 Mother/sister as primary educators 1.3 Male exclusion information 1.4 Early ignorance
Menstrual product accessibility and availability	2.1 High cost and affordability issues 2.2 Distribution and distance 2.3 Shame and embarrassment in purchasing
Physical and Emotional Discomfort	3.1 Pain and fatigue 3.2 Coping strategies 3.3 Emotional toll 3.4 Class abstinences
WASH facilities	4.1 Lack of private changing space 4.2 Limited disposal systems

Table 5. Themes and subthemes of the gendered perspective of MHH.

Lack of early education

Both female and male participants consistently emphasized the absence of structured menstrual education prior to menarche or during early adolescence. The female students shared vivid recollections of fear and confusion upon experiencing menstruation for the first time. One participant reflected, “*When I had my first period, I didn’t know what was happening. I was so afraid*” (F, P-3). Similarly, a male student remarked, “*In rural areas, when girls start menstruating, they know nothing about it. It becomes a crisis*” (M, P-2). This lack of foundational knowledge often leads to misinterpretation of menstrual health outcomes, especially during initial episodes.

Mothers and sisters as primary educators

In the absence of formal education, female students often rely on maternal figures or older sisters for guidance and reassurance. The participants repeatedly mentioned, “*My elder sister understood the matter and told me how to manage it*” (F, P-6), indicating that interpersonal female networks served as the primary knowledge transfer system. In contrast, male participants described this educational transmission as inaccessible to them. One expressed, “*My sister never talked to me about menstruation, even though we’re close*” (M, P-6).

Male exclusion from menstrual information

A deeply entrenched gender divide was evident in access to information. The female participants noted that menstruation was treated as a private matter that must be hidden from men. “*This is something we never say in front of men, it’s shameful*” (F, P-2). Male students confirmed this invisibility, stating, “*We boys never learned anything from girls or family members about menstruation*” (M, P-7). This silence perpetuated myths, misunderstandings, and social distancing.

Early ignorance and social mockery

Ignorance during adolescence contributes to inappropriate behaviors, particularly among boys, including mocking menstruating girls. One male student confessed, “*When we saw stains in school, we laughed because we didn’t understand*” (M, P-2). The female participants shared similar stories of being mocked or misunderstood, exacerbating feelings of shame and isolation during early menstrual experiences: “*The first day I got my period, I didn’t understand what was happening to me...*” (F, P-6).

Menstrual product accessibility and availability

The quantitative disparities in access to menstrual materials by academic program, income, and maternal education (Table 3) are reflected in qualitative accounts describing high costs, limited availability in rural areas, and embarrassment or stigma when purchasing menstrual products.

High cost and affordability issues

Economic barriers were commonly cited by both genders. The female participants noted the financial burden of purchasing sanitary pads: “*It’s quite expensive compared to our income*” (F, P-2). Male participants described affordability related constraints in relation to the continued use of unhygienic alternatives: “*Many women’s still use old cloths because they cannot afford sanitary pads*” (M, P-5). The intersection of poverty and gender norms limits women’s access to safe menstrual materials, even in healthcare student populations.

Distribution and distance

Product accessibility was particularly problematic for rural students. The female participants described having to travel several kilometers to purchase pads: “*It takes me 4 km to get them. The shop is far*” (F, P-1). Male students confirmed this barrier, stating, “*Pads are not always available in villages. Girls have to travel far to buy them*” (M, P-2). Limited on-campus availability compounded the challenge, suggesting a need for institutional support in product distribution.

Shame and embarrassment in purchasing

Shame and embarrassment surrounding menstruation extends into purchasing behavior and shape how female students access menstrual products. Female students expressed reluctance to buy pads in the presence of male

Variables		MHH Indicators												Supporting social environment				Increased discomfort			
		Adequate menstrual materials				Adequate WASH facilities				Adequate menstrual knowledge											
		AOR (95% CI)	P- value	AOR (95% CI)	P- value	AOR (95% CI)	P- value	AOR (95% CI)	P- value	AOR (95% CI)	P- value	AOR (95% CI)	P- value	AOR (95% CI)	P- value	AOR (95% CI)	P- value				
Name of course	Diploma in Midwifery	Ref				Ref															
	Diploma in Nursing & Midwifery	2.96 (0.89–9.8)	0.08			2.72 (1.01–7.32)	<0.05														
	B.Sc in Midwifery (Post Basic)	0.61 (0.21–1.73)	0.36			2.87 (1.28–6.43)	<0.05														
	B.Sc in Nursing (4 years)	6.91 (2.44–19.55)	<0.05			0.64 (0.30–1.38)	0.26														
	MSc in Nursing	2.94 (1.05–8.21)	<0.05			0.88 (0.38–2.02)	0.76														
Year of college	First year	Ref				Ref															
	Second year	0.65 (0.32–1.71)	0.09			2.27 (1.22–4.23)	<0.05														
	Third year	0.67 (0.33–1.87)	0.08			1.92 (1.03–3.57)	<0.05														
	Fourth year	2.18 (1.01–3.31)	<0.05			3.18 (1.04–9.68)	<0.05														
Fathers education level	Graduate or above																				
	Illiterate																				
Mother's education level	Graduate or above	Ref				Ref															
	Higher Secondary	2.27 (0.62–8.31)	0.21			6.19 (1.47–26.00)	<0.05			0.62 (0.19–2.04)	0.44										
	Secondary	4.82 (1.52–15.28)	<0.05			5.06 (1.36–18.85)	<0.05			0.76 (0.36–1.61)	0.48										
	Primary	1.81 (0.59–5.54)	0.29			3.79 (1.01–14.10)	<0.05			1.35 (1.16–3.75)	<0.05										
Birthplace status	Illiterate	2.07 (0.54–7.83)	0.28			5.36 (1.23–23.29)	<0.05			0.97 (0.36–2.60)	0.96										
	Rural									Ref											
	Urban									1.52 (1.30–3.91)	<0.05										
Monthly family income	Below 10,000 BDT																				
	10,000–20,000 BDT																				
	20,000–30,000 BDT																				
	More than 30,000 BDT																				

Table 3. Significant MHH indicators with sociodemographic characteristics in the multivariable logistic regression model. Ref= Reference category; AOR= Adjusted odds ratio; CI= Confidence interval; Both statistically significant and selected non-significant adjusted estimates are shown for completeness.

shopkeepers: “Sometimes I just leave without buying because I feel embarrassed” (F, P-2). Male participants acknowledged this stigma, noting, “Pads are sold in black poly bags, as if it’s something illegal” (M, P-5). This normalization of concealment reinforces embarrassment, restricts women’s autonomy in managing menstruation, and reflects the deep internalization of menstrual taboos within everyday marketplace interactions.

Physical and emotional discomfort

Quantitative evidence of substantial menstrual discomfort and academic disruption (45.4% reported activity disruption and 58.6% reported difficulty participating in class; Table 2) is contextualized by qualitative narratives of pain, fatigue, emotional distress, and coping strategies that often relied on informal support rather than professional care.

Pain and fatigue

Menstrual symptoms such as abdominal pain, backache, and fatigue were universally reported by female participants. “I am in great pain. I have a lot of pain” (F, P-7) and “I feel weak and dizzy” (F, P-3) were typical accounts. Male students also observed these symptoms among peers, describing them as “physically tired and quiet” (M, P-5), reflecting awareness of the somatic burden of menstruation.

Coping strategies

In the absence of professional care, participants adopted informal coping methods. The female students mentioned using hot compresses and painkillers: “Baking massage gives a lot of relief” (F, P-3). Support from peers was also emphasized: “Friends give warm food or stay close” (F, P-4). Male students recognized this dynamic, stating, “Support from friends or family helps them deal with the emotional burden” (M, P-5).

Emotional toll

Mood swings, irritability, and psychological strain were recurrent themes. One female participant said, “I feel very upset. I cannot take good things from people” (F, P-5). Male students noticed these changes, remarking, “Mood swings occur; they become irritated easily and feel mentally disturbed” (F, P-3). Emotional well-being during menstruation emerged as a critical, often overlooked, aspect of MHH.

Class abstinence

Menstrual discomfort often leads to academic disengagement. Female students reported skipping classes and preferring isolation: “I want to rest; I don’t want to do anything” (F, P-7). Male participants acknowledged this, noting, “They often skip class or avoid group work during menstruation” (M, P-5). These disruptions affect educational continuity and highlight the need for menstrual support in academic settings.

WASH facilities

Consistent with the quantitative findings, fewer than half of participants reported that school changing spaces were clean (43.2%) or that they felt safe while changing (47.8%) (Table 2), indicating substantial gaps in WASH adequacy. The qualitative accounts directly explain these quantitative deficits, showing that inadequate privacy, shared/co-gender bathrooms, and weak disposal systems are key reasons why students rated WASH conditions as insufficient.

Lack of private changing space

Inadequate changing facilities are a major barrier to menstrual management. These experiences help contextualize why many students did not perceive school changing spaces as clean or safe in the survey (Table 2), and why WASH adequacy varied by program and year of study in the regression model (Table 3). The female participants reported sharing or cogender bathrooms in hostels and classrooms: “There’s no good changing system... sometimes the same bathroom is used by boys and girls” (F, P-1). Male participants echoed this concern, stating, “There is no proper place to change pads privately” (M, P-3), demonstrating shared recognition of poor infrastructure.

Limited disposal systems

The disposal of used menstrual products has emerged as a critical concern. Female students described wrapping pads in paper and discarding them in unhygienic dustbins: “I throw it in the basket... but it’s dirty” (F, P-7). Male participants discussed unsafe practices such as flushing or open dumping: “In rural areas, they throw used pads in ponds or burn them secretly” (M, P-5). The absence of disposal education and facilities contributes to environmental and health risks.

Discussion

This study provides an in-depth exploration of the gendered dimensions of MHH among nursing students in Bangladesh, illuminating the challenges shaped by gender education, socioeconomic status, and institutional WASH facilities limitations. Despite the participants’ education in health sciences, the findings suggest that both male and female students are deeply embedded within and affected by prevailing sociocultural norms around menstruation. The convergent parallel mixed-methods approach integrating quantitative trends and qualitative narratives yielded a nuanced understanding of how gender intersects with education, economic status, and WASH facilities in relation to MHH indicators.

A striking finding was the substantial proportion of female students who lacked prior knowledge of menstruation before menarche. Nearly half of them were unaware of menstruation before their first period, a finding consistent with previous studies conducted in South Asia and sub-Saharan Africa^{5,30,31}. This paradox high current educational attainment alongside early informational deprivation reflects the timing and source

of menstrual education rather than its absence. Menstrual knowledge was transmitted primarily through mothers or older female relatives, often reactively at menarche and framed through secrecy and restriction. This familial transmission of knowledge, although well-intentioned, may perpetuate misinformation and secrecy, as it lacks scientific grounding and is constrained by generational taboos^{14,32}. These findings highlight the need for pre-menarche, curriculum-embedded menstrual education that is medically accurate and gender inclusive, extending into pre-service nursing training.

Male students' narratives revealed near-total exclusion from menstrual education, which reinforced early mockery and discomfort. However, qualitative findings also showed a transition from ignorance to empathy during professional training. These accounts echo findings from earlier qualitative studies in India, which revealed that boys are socialized to avoid menstruation-related topics and that their exclusion reinforces menstrual stigma and shame^{33–35}. The silence around menstruation, especially in coeducational contexts, not only isolates girls but also prevents boys from becoming informed allies. This gendered knowledge gap may have long-term implications for future clinical practice, particularly when these individuals become healthcare providers responsible for delivering sexual and reproductive health services.

Economic barriers were another salient issue and a majority of students reported access to menstrual materials, disaggregated analysis revealed persistent inequities. Students from higher-income households could maintain consistent use of preferred products and manage pain and hygiene privately, whereas lower-income students could only intermittently purchase, lower-quality products, mirroring existing evidence that socioeconomic privilege improves access to health commodities^{36,37}. The apparent contradiction of high overall access but strong income effects can be explained by conditional affordability, often resorting to unhygienic alternatives, a pattern also reported in studies from Ethiopia and Bangladesh^{5,13,38}. The qualitative findings further emphasize the emotional toll associated with purchasing menstrual products, as many female participants expressed reluctance or embarrassment when buying pads from male shopkeepers. Some even reported abandoning the purchase altogether due to discomfort, an insight supported by studies on menstrual taboos and marketplace stigma^{8,39}. Together, the quantitative income gradients observed in the regression models and the qualitative narratives of embarrassment during purchasing illustrate how social stigma amplifies the effects of financial hardship, resulting in reduced access to safe menstrual products among lower-income students.

Menstrual pain, fatigue, and emotional instability were consistently reported by female participants, underscoring the physical and psychological burden of menstruation. Half of the students managed their menstrual pain, with many students reporting using home remedies such as hot compresses or taking painkillers without medical consultation in the qualitative findings. This duality of access to medical knowledge yet reliance on informal coping strategies suggests that even among educated populations, menstruation remains a private and often unspoken struggle^{4,37}. While being physically unaffected, male students demonstrated a growing awareness of their female peers' distress, a shift that indicates the potential for transforming traditional masculine attitudes through inclusive education¹⁸. These findings highlight that despite medical training, menstruation remains a private and often unsupported experience, though male students' increasing empathy offers hope for more inclusive care.

Inadequate WASH facilities emerged as a cross-cutting barrier to effective MHH management. Fewer than half of the students found their school's changing facilities clean, and qualitative testimonies revealed an alarming lack of privacy in hostels and classrooms. Shared bathrooms between male and female students, inadequate disposal bins, and the absence of designated changing spaces have emerged as serious concerns echoing global findings that poor WASH infrastructure undermines safe menstrual management and school attendance^{14,38}. Notably, both male and female students highlighted the lack of disposal education and facilities, with some rural participants reporting the practice of burning or discarding used pads in open environments. These practices not only threaten environmental safety but also reflect institutional negligence in integrating menstrual management needs into campus design and maintenance.

Parental education, particularly maternal literacy, was found to significantly association with students' MHH practices and perceptions. The finding that students whose mothers had secondary education showed higher odds of adequate menstrual materials compared with those whose mothers had graduate-level education, as it may reflect unmeasured contextual factors or the small number of mothers with graduate-level education in the sample. This finding is consistent with prior research showing that maternal literacy enhances daughters' menstrual health practices, knowledge, and emotional coping^{16,40}. The qualitative data revealed that students from rural areas were less likely to possess adequate menstrual knowledge, reinforcing the rural–urban divide in terms of access to health education and commodities. Monthly income remained a strong protective factor against discomfort, with higher-income students reporting fewer menstrual disruptions, suggesting that economic stability enables access not only to products but also to comfort-enhancing environments. Maternal education, rural background, and family income were significantly association with nursing students' menstrual health experiences and access to essential resources.

Practical implications for institutions

Fourth-year students showed higher odds of reporting access to adequate menstrual materials, which may reflect cumulative educational exposure and increased familiarity with MHH over time. Differences by year of study may also reflect reporting bias, residual confounding (such as prior menstrual education or individual coping strategies), or institutional variation in facilities and support across academic years, rather than educational progression alone. For the nursing colleges studied, the findings highlight several actionable priorities. First, integrating comprehensive, gender-inclusive MHH content into nursing curricula can reduce stigma, improve peer support, and better prepare future healthcare providers. Second, institutions should ensure minimum MHH standards in campus WASH facilities, including private changing spaces, reliable water access, disposal bins, and routine maintenance. Third, student support mechanisms such as access to affordable menstrual

products or referral pathways for menstrual pain management can directly improve attendance, comfort, and learning outcomes.

Policy and government implications

At the policy level, the results call for the integration of MHH into national education standards, overseen by relevant government bodies. Investment in gender-responsive WASH infrastructure within public educational institutions is essential and aligns with broader commitments to gender equity, adolescent and young adult health, and sustainable sanitation. Additionally, government-led guidelines on menstrual waste management in educational settings are needed to address environmental risks and institutional gaps. Strengthening intersectoral collaboration between health, education, and sanitation authorities would ensure that MHH is treated not as a private issue but as a core component of public health and professional training.

Public health implications

This study highlights MHH as a critical public health concern among nursing and midwifery students, who are both a vulnerable group and future healthcare providers. Persistent gaps in menstrual knowledge, stigma, and inadequate access to products and WASH facilities may negatively affect students' wellbeing, academic engagement, and future capacity to deliver sexual and reproductive health care. Integrating gender-inclusive MHH education into pre-service nursing curricula and improving gender-responsive WASH infrastructure within educational institutions are essential public health interventions. Addressing socioeconomic and rural-urban disparities through equity-focused strategies can further reduce preventable menstrual inequities and strengthen health system capacity. Together, the quantitative associations and qualitative narratives indicate that improving MHH in nursing education requires a dual strategy: transforming gender norms through inclusive learning and removing institutional barriers through menstrual-responsive services and infrastructure.

Limitations

This study has several limitations that should be acknowledged. First, the research was conducted exclusively among students from government nursing institutions, which may limit the generalizability of the findings to private or non-governmental settings. However, the inclusion of seven diverse public institutions across different geographic regions of Bangladesh aimed to enhance the representativeness of the sample within the public sector context. Second, the cross-sectional design of the study limits causal inference. To mitigate this, the study employed a convergent mixed-methods approach, combining quantitative data with qualitative insights to strengthen the depth and contextual understanding of the findings. Third, the reliance on self-reported data and a substantial proportion of "do not remember" responses were observed for several survey items, indicating the possibility of recall bias and social desirability bias. To reduce this risk, participants were assured of confidentiality and anonymity, and the survey and FGDs were facilitated by trained female researchers with experience in conducting sensitive health-related interviews. Fourth, while gender-segregated FGDs encouraged open discussion and helped participants feel more comfortable, it is possible that group dynamics, peer influence, or institutional hierarchies may have influenced individual responses. To address this, facilitators used culturally sensitive and nonjudgmental prompts, ensured equitable participation, and conducted debriefings to reflect on potential bias in group interactions.

Conclusion

This mixed-methods study shows that MHH among nursing and midwifery students in Bangladesh is shaped by intersecting gaps in menstrual education, economic constraints, and inadequate institutional WASH support. Despite being health trainees, many students, particularly those from rural and lower-income backgrounds and those with less-educated mothers, reported limited early menstrual knowledge, limited access to menstrual materials, and insufficient privacy and disposal options on campus. Importantly, the inclusion of male students' perspectives revealed that menstruation remains framed by silence and shame, although professional training and shared academic environments can foster greater awareness.

These findings extend existing evidence beyond adolescent populations by highlighting menstruation as both a gendered and systems-level challenge within nursing education. To promote menstrual dignity and equitable health training, nursing institutions should: (1) integrate structured, gender-inclusive MHH content into pre-service curricula; (2) ensure minimum WASH standards, including private changing spaces, reliable water, disposal bins, and routine maintenance; and (3) establish practical student supports such as affordable product access and clear referral pathways for menstrual pain management. At the policy level, coordinated action between education, health, and sanitation authorities is needed to strengthen implementation, monitoring, and menstrual waste management guidance across public training institutions.

Data availability

The datasets generated and/or analysed during the current study are not publicly available due to the sensitive nature of the topic and concerns regarding participant privacy, but are available from the corresponding author on reasonable request.

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Author contributions

****M.N.U.Z.**** conducted the conceptualization, methodology design, investigation, formal analysis, data curation, software, resource management, and wrote the original draft. ****A.S.**** contributed to conceptualization, supervision, project administration, resource support, and manuscript review and editing. ****T.Y.**** was responsible for conceptualization, methodology, supervision, project administration, funding acquisition, and manuscript review and editing. All authors reviewed and approved the final manuscript.

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Competing interests

The authors declare no competing interests.

Consent for publication

All authors consent for publication.

Additional information

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